

Testing Plan

Sprint 3

FIT2101 Team 5

Introduction:

The testing plan being conducted will be focused on the tasks allocated to group members in Sprint 3. This testing plan will document each task's requirements and input/output expected of each task. This testing plan will also clearly outline how the test can be easily reproduced by the user.

User stories:

This lists all user stories that will be incorporated into the tasks created, giving reasoning for tasks that were created and enacted upon:

THEME: Task Popup:

- As a user, I want to be able to log the amount of time I spent on a task, so that I can keep track of how much time I and my team have spent on each task.
- As a user, I want to be able to pick a date using a calendar so that it is easy to select a date.
- As a user, I want to be able to delete tasks so that I can properly manage my project's tasks.

THEME: Sprints

- As a user, I want the statuses to be colour-coded so that I can differentiate between statuses visually.
- As a user, I want to be able to see the sprint tasks in the form of a Kanban board so that I can easily see what tasks are not started, in progress and completed.
- As a user, I want to be able to see a burndown chart on the Sprint Board to help track how the team is progressing through the sprint.
- As a Scrum Master, I want to be able to delete sprints so that I can correct mistakes or remove completed sprints from the list.

Testing process:

The testing processes that will be used to test new implementations on the website will be primarily based on functional testing and UI testing. This is due to the tasks of the sprints being based on new functionality implementation and the ease of navigation for the user to perform tasks. Tasks are labelled with the testing process they will be conducted within the task testing analysis portion below.

Task testing analysis:

THEME: Task Popup:

Logging Time:

When working on a task, we want to keep track of the time spent on the task by me or the team. This makes it easy to check the work velocity and the time taken on each task.

UI testing: When logging in time in a particular task, it should show the assignee list as a drop-down menu and the user can select the team member and log in the time.

Functionality testing: Once the user logs in his/her time, the burndown chart and the team management graph should reflect the corresponding time log. Also, when clicking on a task to view details, it should show logged in time.

Recreate test:

1. Click on a pre-existing task (if no tasks are present, create one)
2. Click on the edit button to edit details of the task
3. Click the 'plus' button beside the time log field
4. Select user from the drop-down list
5. Enter the date or select it from the date picker
6. Add in time in hours in the empty time field
7. Time should be reflected on the task detail's view

Logging Time – Date Picker:

When logging in time, a user can either enter the date manually or use a date picker to select the date of time log. This makes picking date easier without having to worry about the correct date format to be entered.

UI testing: When logging in time, a calendar icon must be visible near the date field to select dates

Functionality testing: The date picker must not allow users to select a future date to log in time

Recreate test:

1. Click on a pre-existing task (if no tasks are present, create one)
2. Click on the edit button to edit details of the task
3. Click the 'plus' button beside the time log field
4. Select user from the drop-down list
5. Using the calendar icon on the date field, select a date for the time log.
6. Date should show up in the correct format once selected from the date picker

Deleting Tasks:

As a user we might need to edit tasks or delete them entirely from the product backlog. So, we would want the option to delete tasks that are not required anymore.

UI testing: The delete task icon is highlighted in blue with a 'bin' image in it.

Functionality testing: Once a task is deleted, it will be removed from the product backlog

Recreate test:

1. Open the product backlog page
2. If there are no tasks in the product backlog, create one
3. Click on the delete task icon to delete the task
4. The task should be removed from the product backlog

THEME: Sprints:

Status Colours:

When a user is on the 'Sprints' page, the status of each sprint is shown. Different status colours for different stages of a sprint make it easier to distinguish the current position of a sprint.

UI testing: When a status is selected, the status is highlighted in different colours such as: not started is 'red', in progress is 'pink' and completed is 'green'.

Functionality testing: When the status of a sprint changes, the highlighted status colour of the sprint must also change to the according hue.

Recreate test:

1. Click on 'Sprints' page on the left-hand side of the UI
2. If there are no sprints present, create one
3. Set status of the sprint to not started and the status colour must be 'red'
4. After starting the sprint, the status must change to in progress and the highlighted colour must be 'pink'
5. Next, end the sprint and the status changes to completed and highlighted in 'green' colour

Kanban Board:

A Kanban board is an agile project management tool designed to help visualise work, limit work-in-progress, and maximise efficiency (or flow). Having a Kanban board allows the user to see all the tasks in a broader way and in a well-defined table kind of format.

UI testing: Tasks displayed in a Board like format with each task in its respective column

Functionality testing: When adding tasks to a sprint, the task must be displayed in the Kanban board. Whenever the status of a task changes, the task must move over to its corresponding column in the Kanban board

Recreate test:

1. Open a sprint containing tasks, if not, create a sprint and add tasks to it.
2. When the sprint is opened, the Kanban board must contain all the tasks in that sprint.
3. Click on any of the tasks in the Kanban board
4. Edit the task and change its status to a different one from the status
5. The task moves to its corresponding location in the Kanban board

Burndown Chart:

A burndown chart shows the progression of a sprint in a graph format. It shows the user the ideal velocity that should be maintained to finish the sprint on time, the actual velocity of the team during the sprint and the hours left before the sprint duration ends. All this information is very helpful to keep track of the progress and can help increase or decrease the velocity of the sprint.

UI testing: The graph must have 3 lines in different colours to show the necessary information in the chart. There is a legend to show which line represents what value

Functionality testing: After starting a sprint, the burndown chart shows up in the sprint details and keeps updating as time goes on.

Recreate test:

1. Click on an existing sprint or a new sprint
2. If the sprint has not been started yet, start the sprint.
3. The burndown chart will show up at below the Kanban board, with graph lines displaying data like actual velocity, ideal velocity and hours left for the sprint to end

Deleting Sprints:

Completed or non-required sprints will add up in the 'Sprints' page. As a solution for this, a scrum master may choose to delete completed sprints or sprints that no longer serve a purpose to the project.

UI Testing: The delete icon is a 'blue' highlighted icon with a 'bin' image in it. It is present in the 'Sprints' page, beside each sprint.

Functionality Testing: When a sprint is deleted, it must be removed from the 'Sprints' page and the tasks in the sprint are moved back to the product backlog.

Recreate test:

1. Click on 'Sprints' page
2. If there are no existing sprints, create a sprint
3. Each sprint in the 'Sprints' page must have the delete button near the status tab
4. Click on the sprint
5. Check if it has any tasks in it (make a note if yes or no)
6. Go back to 'Sprints' page
7. Click on 'delete' button to delete the sprint
8. If the deleted sprint had any tasks in it, go back to product backlog
9. The product backlog must display the tasks from the deleted sprint (if any)

Bug Fixing:

This list covers all bugs that were fixed during Sprint 3. These were either identified during Sprint 2 Review or via internal testing. Each item in the list will specify how the bug was affecting functionality prior to Sprint 2, and how it should actually work post Sprint 3.

BUG: Story Name requirement when creating a task; Story Name could be left blank during task creation, even though it is a required field during task creation.

UI Testing: When the Story Name input field is empty, the input box should be highlighted red. There should be text underneath the box stating: "Story must have a name". The "Save" button should be greyed out when the Story Name input field is empty.

Functionality Testing: It should not be possible to create a task without filling the Story Name input field, i.e. the greyed out 'Save' button should not be clickable.

Recreate Test:

1. Navigate to the Product Backlog Page by clicking on the "Product Backlog" button in the navigation bar on the left side of the screen.
2. Enter task creation by clicking the plus button on the top right corner of the screen.
3. Fill in task information as required.
4. Note that the Story Name input field is initialised as blank.
5. Note that the Story Name input field is highlighted red, with text underneath reading "Story must have a name".
6. Note that the "Save" button in the top right corner of the popup is greyed out.
7. Try to click the "Save" button and note that nothing is happening as the Story Name field is still blank.
8. Now enter text into the Story Name input field.
9. The Story Name input field is no longer highlighted red, and the "Save" button is no longer greyed out.
10. It is now possible to create a task.

BUG: Story Points requirement when creating a task; Story points could have been entered as any non-numeric string rather than simply a number.

UI Testing: The Story Points input field during task creation should now be a numeric input field with arrows, allowing the user to increase or decrease the number of story points per task.

Functionality Testing: The Story Points input field should no longer allow the user to type directly into the field, it should allow the user to increase/decrease the number of story points with the use of arrow buttons inside the input field.

Recreate Test:

1. Navigate to the Product Backlog Page by clicking on the “Product Backlog” button in the navigation bar on the left side of the screen.
2. Enter task creation by clicking the plus button on the top right corner of the screen.
3. Note that the user is now unable to type directly into the Story Points input field.
4. Increase/decrease the number of Story Points with the arrows inside the input field.
5. Fill out other information as required.
6. Save the task with the blue button on the top right corner of the popup.
7. Verify that the correct number of story points was allocated.

BUG: Specifying Sprint Date requirement during creation; the Sprint date range could allow any date range, e.g. Creating Sprints that start before the current date, invalid date ranges, etc.

UI Testing: When selecting the date range of a Sprint, all dates prior to the current date should be greyed out in the Calendar Popup. When selecting the end date of a Sprint, all dates prior to the end date should also be greyed out.

Functionality Testing: The calendar popup should not allow a user to input an invalid date, i.e. sprint creation should not be possible without a valid date range.

Note that testing was completed on the 17/10/2022, please adjust the recreation of the test accordingly.

Recreate Test:

1. Navigate to the Product Backlog Page by clicking on the “Sprints” button in the navigation bar on the left side of the screen.
2. Enter sprint creation by clicking the plus button on the top right corner of the screen.
3. Enter a sprint name.
4. Enter a sprint start date by clicking on the calendar icon in the Sprint Start Date input field.
5. Note that in the calendar popup, all dates prior to 17/10/2022 have been greyed out, disallowing the user to enter such dates.
6. Now select a sprint start date in the future, e.g. 24/10/2022

7. Next, click the calendar icon inside the Sprint End Date input field.
8. Note that all dates prior to 24/10/2022 (Sprint Start Date) have been greyed out.
9. Now, select a sprint end date after the sprint start date, e.g. 30/10/2022.
10. Verify a sprint has been created with the specified start and end dates.

BUG: Resetting main page; Clicking on same page doesn't bring user to main screen

This bug refers to the situation where, for example, a user would have clicked 'into' a task on the 'Product Backlog' page to check the task information, but would be unable to return back to the main 'Product Backlog' page by clicking the 'Product Backlog' button on the navigation bar. The same issue would occur with individual Sprints and the main 'Sprints' page.

Functionality Testing: The user should be able to return to the main page by clicking on the buttons on the navigation bar.

Recreate Test:

1. Create a dummy task and sprint prior to testing. Please refer to Sprint 2 Testing Plan if required.
2. Navigate to the 'Product Backlog' page via the navigation bar on the left side of the screen.
3. Inspect the dummy task information by clicking on the task.
4. Now click on the 'Product Backlog' button on the navigation bar.
5. Verify that the screen no longer displays task information, and now shows the 'Product Backlog' main page.
6. Navigate to the 'Sprints' page via the navigation bar on the left side of the screen.
7. Inspect the dummy sprint information by clicking on the sprint.
8. Now click on the 'Sprints' button on the navigation bar.
9. Verify that the screen no longer displays sprint information, and now shows the 'Sprints' main page.